

Social Determinants of Health (SDoH) Extraction from Clinical Notes Using NLP

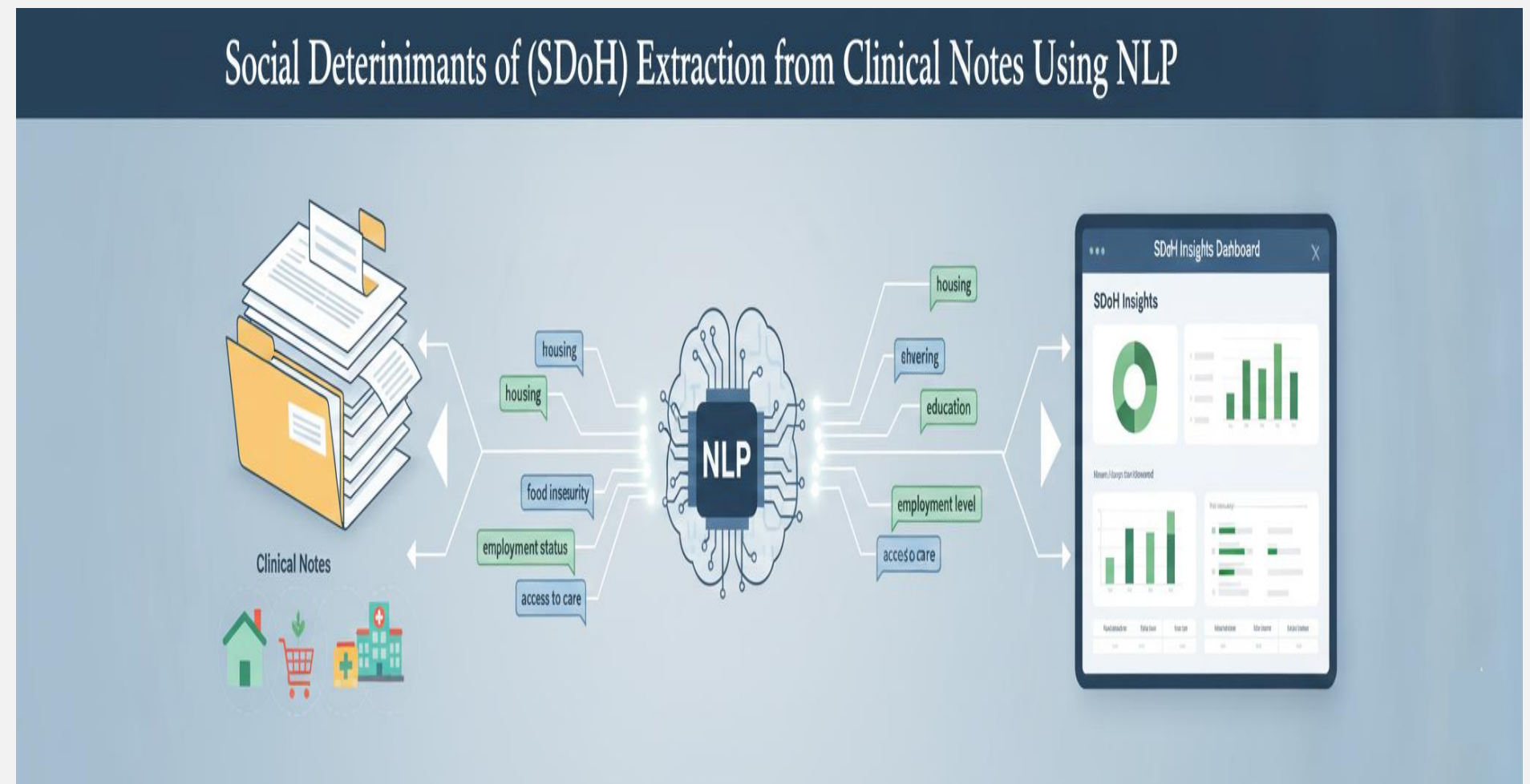
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Introduction

Understanding the Challenge

SDoH strongly influence patient outcomes.

But these factors are hidden inside
unstructured EHR clinical notes.

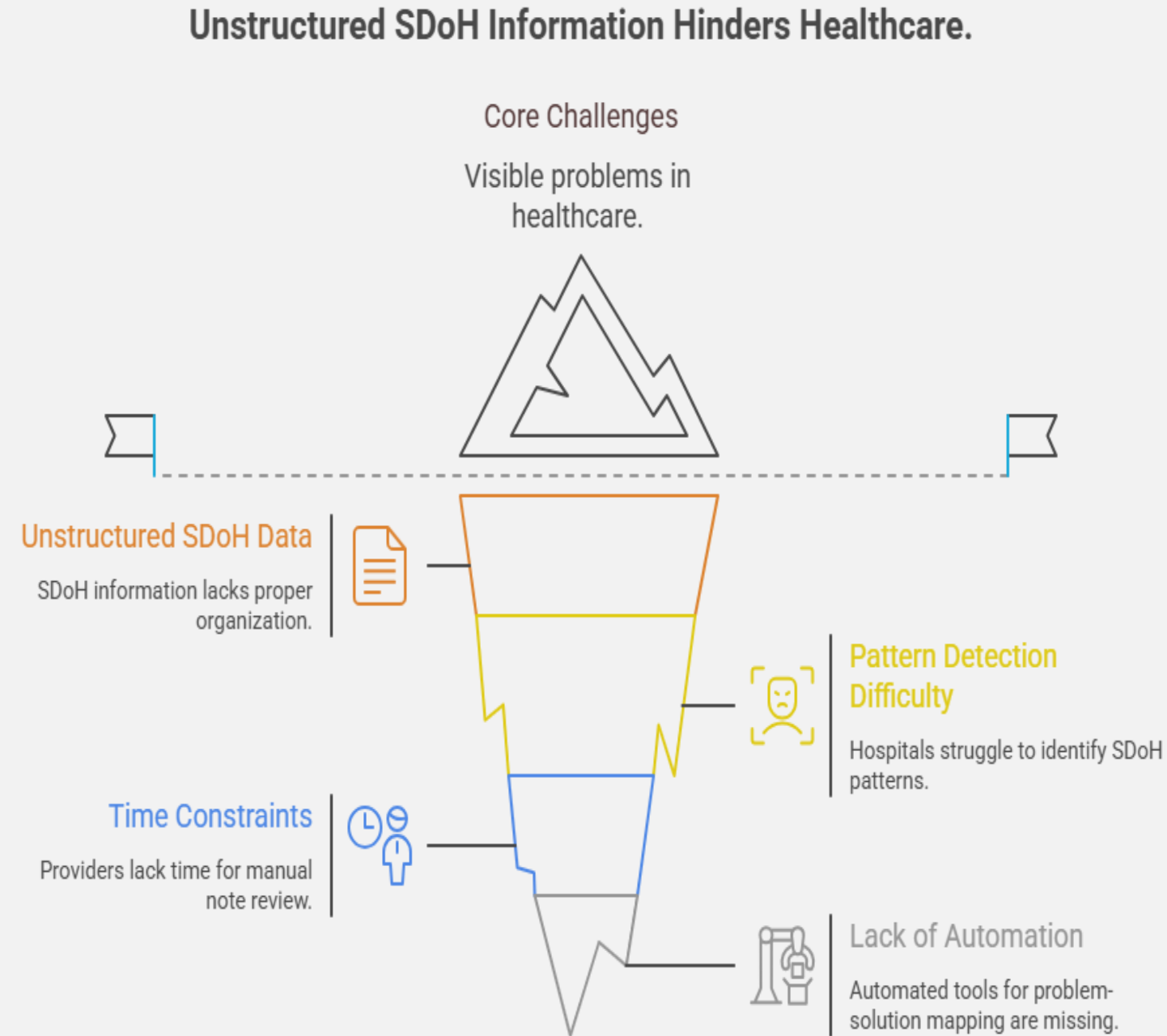
Manual extraction is time-consuming.

Our project uses **NLP + Rule-Based detection**
to extract SDoH categories automatically.

Problem Statement

The Core Challenges

- SDoH information is not structured.
- Difficult for hospitals to detect patterns.
- Providers do not have time to manually read long notes.
- Lack of automated tools to map problems → solutions.



Aim of the Project

Build an **automated tool** that:

Reads clinical notes

Detects SDoH issues using keyword rules

Instantly provides **solutions & resources**

Helps healthcare workers support patients quickly.

Literature Review (Summary)

Research Foundation



Medical image segmentation using multimodal prompts (SAT model).



AI-based clinical decision support concepts (Algorithmic Consultant).



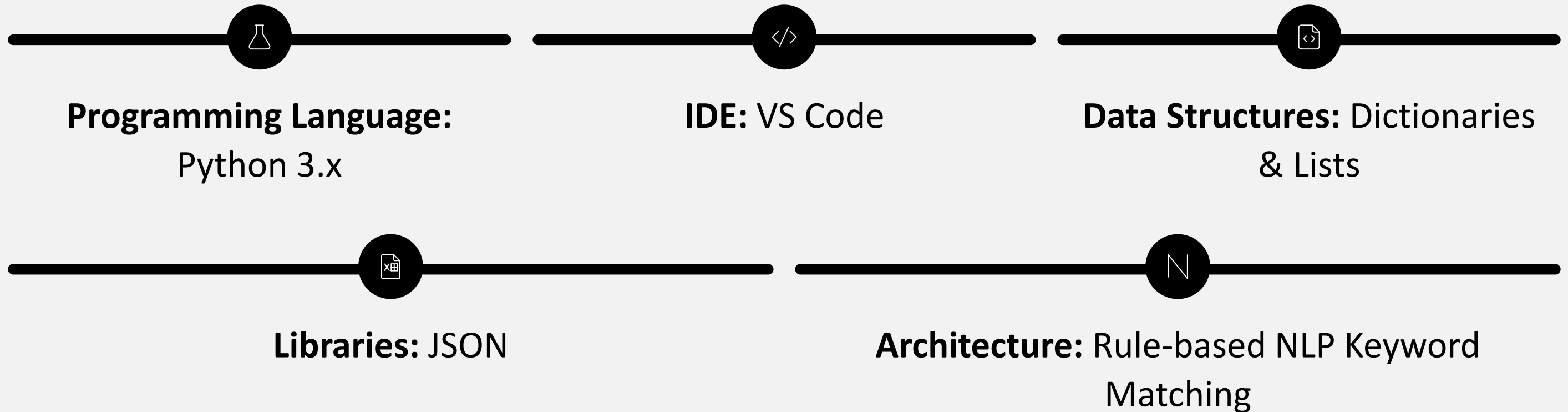
Predictive models for opioid usage after surgery.

Data governance models for healthcare data.

SDoH extraction research using LLaMA, ClinicalBERT, etc.

Tools & Techniques

Technical Stack



Clinical Note Processing Flow



Input Clinical Note



Normalise Text



Keyword Matching



Detect SDOH Category



Fetch Recommended Solutions



Output Report and JSON

System Architecture

Processing Flow



Input clinical note



Normalise text



Keyword matching



Detect SDoH category



Fetch recommended solutions



Output report + JSON

Knowledge Base (Rules_solution.py)

Comprehensive SDoH Coverage

Consists of **21 SDoH categories**

Covers Housing, Food, Employment,
Transportation, Mental Health, etc.

Each category has:

- Keywords
- Solutions/Resources

Methodology ([Main.py](#))

Core Functions

1

`detect_sdoh_from_clinical_notes()`

2

`get_solutions()`

3

`generate_json_output()`

4

`print_report()`

Result

Sample Output (Case 1)

Detected: Housing, Employment, Transportation

Solutions Provided:

- National Housing Hotline
- Unemployment benefits
- Public transport resources

Sample Output (Case 2 & 3)

Case 2: Social Support, Neighbourhood

Case 3: Employment, Social Support, Tobacco Use

System accurately identifies social issues from notes.

Converts unstructured data → structured SDoH report.

Provides immediate actionable resources.

Improves screening speed & reduces manual work.

Future Scope

- Add machine learning / deep learning models
- Integrate with EHR systems
- Add database storage
- Add dashboards & visualisation
- Multilingual support
- Mobile app integration
- Real-time hospital deployment

Conclusion

Prototype proves rule-based SDoH extraction is effective. Helps clinicians quickly understand patient social issues. Makes referral process faster and more efficient

THANK YOU